

MethylPipeline: A Scalable Generative Framework for Whole-Genome Bisulfite Sequencing-Based Disease Classification

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Abstract

Whole-genome bisulfite sequencing (WGBS) enables base-resolution profiling of DNA methylation, but the scale and complexity of these data pose substantial computational and statistical challenges for biomarker discovery and disease classification. Here we present *MethylPipeline*, an end-to-end, GPU-accelerated analytical framework that performs (i) alignment and methylation extraction, (ii) group-wise centroid construction, (iii) differential methylation analysis, (iv) genomic mapping and enrichment, and (v) generative classification. The pipeline is designed around a per-chromosome, per-context architecture and relies on aggregated group-level objects, *MethylBetaCentroids* and *MethylBetaMixtureCentroids*, which enable scalable modeling of methylation distributions without storing individual-level matrices. Final classification is based on a weighted log-likelihood ratio derived from position-specific Beta Mixture Models and biologically informed weights, avoiding generic machine learning models that discard distributional information. The resulting framework is flexible, interpretable, and suitable for large-scale WGBS studies aiming at early disease detection and robust methylation biomarker panels.

1 Introduction

DNA methylation is a key epigenetic modification implicated in gene regulation, development, and disease. Whole-genome bisulfite sequencing (WGBS) provides base-pair resolution of methylation states but generates data sets with billions of reads and tens of millions of cytosine sites. Extracting reliable biomarkers and building predictive models in this setting requires both computational scalability and statistically principled methods.

Traditional pipelines often rely on per-sample matrices of methylation levels and subsequently apply point-estimation-based tests or generic machine learning algorithms. These approaches (i) struggle with memory and I/O constraints at large scale, (ii) often ignore multimodality in methylation distributions, and (iii) discard substantial information about the underlying distributions when compressing data to simple statistics.

MethylPipeline addresses these limitations by:

- leveraging GPU-accelerated alignment,
- representing group-level methylation via centroids aggregated per chromosome and context,
- modeling key cytosine sites with Beta Mixture Models (BMMs), and
- using a generative, weighted log-likelihood classifier that respects the full distributional structure.

In what follows, we describe the full computational pipeline and provide formal definitions of the statistical components, with emphasis on the final generative classifier.

2 Methods

2.1 Overview of the MethylPipeline Workflow

The analytical workflow consists of eight main stages:

1. GPU-accelerated alignment of WGBS reads,
2. methylation extraction,
3. centroid construction,
4. initial detection of differentially methylated positions (DMPs),
5. genomic mapping and enrichment,
6. refinement via Beta Mixture Models (BMMs),
7. final DMP evaluation, and
8. generative classification using weighted likelihood aggregation.

All stages are executed independently by chromosome c and cytosine context $t \in \{\text{CG}, \text{CHG}, \text{CHH}\}$, and the results are aggregated in the final classification step.

We denote groups by $g \in \{\text{Healthy}, \text{Disease}\}$ throughout.

2.2 Alignment and Methylation Extraction

2.2.1 GPU-Accelerated Alignment

Whole-genome bisulfite FASTQ files are aligned using NVIDIA Clara Parabricks or an equivalent GPU-accelerated alignment framework. The standard workflow includes:

- read alignment,
- duplicate marking,
- base quality score recalibration (BQSR), and
- coordinate sorting.

The output of this stage is a set of BAM files per sample, with high consistency across groups.

2.2.2 Extraction of Methylated and Unmethylated Cytosine Counts

Each BAM file is processed by a module referred to as *MethylExtractor*. For each chromosome c and context t , an HDF5 file is generated containing:

- genomic position p ,
- methylated counts $mC_{i,p}$,

- unmethylated counts $uC_{i,p}$,
- trinucleotide context and optional quality metrics.

For sample i at position p , the methylation level is defined as:

$$y_{i,p} = \frac{mC_{i,p}}{mC_{i,p} + uC_{i,p}}, \quad (1)$$

whenever $mC_{i,p} + uC_{i,p} > 0$; positions with zero coverage can be handled via imputation or exclusion.

2.3 Group-Level Centroids: MethylBetaCentroids

Let $g \in \{\text{Healthy}, \text{Disease}\}$ denote the group. For each group, chromosome, and context, we construct a *MethylBetaCentroid* that aggregates methylation information across individuals.

For each position p , the centroid stores:

$$N_{g,p} = \text{number of samples in group } g \text{ with valid coverage at } p, \quad (2)$$

$$S_{g,p}^{(1)} = \sum_{i=1}^{N_{g,p}} y_{i,p}, \quad (3)$$

$$S_{g,p}^{(2)} = \sum_{i=1}^{N_{g,p}} y_{i,p}^2, \quad (4)$$

$$\Lambda_{g,p}^{(x)} = \sum_{i=1}^{N_{g,p}} \log(y_{i,p} + \varepsilon), \quad (5)$$

$$\Lambda_{g,p}^{(1-x)} = \sum_{i=1}^{N_{g,p}} \log(1 - y_{i,p} + \varepsilon), \quad (6)$$

where $\varepsilon > 0$ is a small constant to avoid numerical issues at 0 and 1.

The empirical mean and variance for group g at position p are:

$$\bar{y}_{g,p} = \frac{S_{g,p}^{(1)}}{N_{g,p}}, \quad (7)$$

$$s_{g,p}^2 = \frac{S_{g,p}^{(2)}}{N_{g,p}} - \bar{y}_{g,p}^2. \quad (8)$$

These statistics enable fast estimation of Beta parameters and computation of divergences without storing individual-level data.

2.4 Initial Detection of Differentially Methylated Positions (DMPs)

For each chromosome and context block, a *MethylModeler* step compares group centroids and identifies preliminary DMPs.

2.4.1 Effect Size

We define the difference in group means:

$$\Delta\mu_p = \bar{y}_{\text{Disease},p} - \bar{y}_{\text{Healthy},p}. \quad (9)$$

A pooled variance estimate is:

$$\sigma_p^2 = \frac{(N_{\text{Healthy},p} - 1) s_{\text{Healthy},p}^2 + (N_{\text{Disease},p} - 1) s_{\text{Disease},p}^2}{N_{\text{Healthy},p} + N_{\text{Disease},p} - 2}. \quad (10)$$

These summaries can be used in standard test statistics or integrated into more complex divergence metrics.

2.4.2 Divergence Measures

Let $f_{\text{Healthy},p}(y)$ and $f_{\text{Disease},p}(y)$ denote fitted Beta distributions for each group at position p . MethylPipeline can employ multiple divergence measures, including:

Squared Hellinger distance

$$H^2(p) = 1 - \int_0^1 \sqrt{f_{\text{Healthy},p}(y) f_{\text{Disease},p}(y)} dy. \quad (11)$$

Jensen–Shannon divergence Define the mixture $m_p(y) = \frac{1}{2}f_{\text{Healthy},p}(y) + \frac{1}{2}f_{\text{Disease},p}(y)$. The Jensen–Shannon divergence is:

$$\text{JS}(p) = \frac{1}{2}D_{\text{KL}}(f_{\text{Healthy},p} \parallel m_p) + \frac{1}{2}D_{\text{KL}}(f_{\text{Disease},p} \parallel m_p), \quad (12)$$

where $D_{\text{KL}}(f \parallel g) = \int_0^1 f(y) \log\{f(y)/g(y)\} dy$ is the Kullback–Leibler divergence.

Positions exceeding user-defined thresholds in effect size and divergence form a preliminary DMP set:

$$D^{(0)} = \bigcup_{c,t} D_{c,t}^{(0)}, \quad (13)$$

where $D_{c,t}^{(0)}$ is the set of preliminary DMPs on chromosome c and context t .

2.5 Genomic Mapping and Biological Enrichment

The preliminary DMPs in $D^{(0)}$ are mapped to genomic features such as promoters, gene bodies, exons, introns, and regulatory regions. Let $w_p^{(\text{bio})}$ denote a biological weight for position p , encapsulating region type (e.g., promoter, exon), gene-level importance (e.g., known disease genes), and pathway enrichment.

This yields a biologically enriched DMP list annotated with gene mappings and functional information.

2.6 Refinement via Beta Mixture Models (MethylBetaMixtureCentroids)

To capture multimodality within each group, the shortlisted positions in $D^{(0)}$ undergo Beta Mixture Model (BMM) fitting.

2.6.1 Beta Mixture Model

For group g and position p , we fit a mixture model:

$$f_{g,p}(y) = \sum_{k=1}^{K_p} \pi_{g,p,k} \text{Beta}(y \mid \alpha_{g,p,k}, \beta_{g,p,k}), \quad (14)$$

where:

- $K_p \in \{1, 2, 3\}$ is the number of mixture components chosen via information criteria (e.g., AIC, BIC),
- $\pi_{g,p,k} \geq 0$ and $\sum_{k=1}^{K_p} \pi_{g,p,k} = 1$,
- $\alpha_{g,p,k} > 0$ and $\beta_{g,p,k} > 0$ are shape parameters of the Beta distribution.

The fitted parameters for all positions form the *MethylBetaMixtureCentroid* for each group.

2.6.2 Mixture-Based Divergence

To compare groups at position p , we can compute a symmetric KL divergence between mixtures:

$$D_{\text{SKL}}(p) = D_{\text{KL}}(f_{\text{Healthy},p} \parallel f_{\text{Disease},p}) + D_{\text{KL}}(f_{\text{Disease},p} \parallel f_{\text{Healthy},p}). \quad (15)$$

Other mixture-based divergences (e.g., Jensen–Shannon) may also be used.

Positions that remain strongly differentiated under the mixture regime define the final DMP set:

$$D = \bigcup_{c,t} D_{c,t}, \quad (16)$$

where $D_{c,t} \subseteq D_{c,t}^{(0)}$ is the final DMP set for chromosome c , context t .

2.7 Generative Classification via Weighted Log-Likelihood Aggregation

Instead of fitting a discriminative model (e.g., random forest, XGBoost) on summary features, MethylPipeline implements a generative classifier that uses the per-position Beta Mixture Models and biologically/statistically informed position weights.

2.7.1 Per-Position Likelihood

For a sample i , let $y_{i,p}$ denote its methylation level at position p [cf. (??)]. The likelihood of $y_{i,p}$ under group g is:

$$p(y_{i,p} \mid g, p) = \sum_{k=1}^{K_p} \pi_{g,p,k} \text{Beta}(y_{i,p} \mid \alpha_{g,p,k}, \beta_{g,p,k}), \quad (17)$$

where the mixture parameters are given by (??).

For numerical stability, we work with the log-likelihood:

$$\ell_{g,p}(y_{i,p}) = \log(p(y_{i,p} \mid g, p) + \varepsilon), \quad (18)$$

where $\varepsilon > 0$ is a small constant to avoid log 0 in edge cases.

2.7.2 Position-Specific Weights

Each position $p \in D$ is assigned a total weight w_p that integrates multiple aspects:

$$w_p = w_p^{(\text{stat})} w_p^{(\text{bio})} w_p^{(\text{div})} w_p^{(\text{cov})}, \quad (19)$$

where, for example:

- $w_p^{(\text{stat})}$ encodes statistical strength (e.g., effect size, inverse of q -value),
- $w_p^{(\text{bio})}$ encodes region type and gene-level importance,
- $w_p^{(\text{div})}$ encodes magnitude of divergence between groups,
- $w_p^{(\text{cov})}$ encodes coverage reliability and technical robustness.

Weights are normalized such that:

$$w_p \geq 0, \quad \sum_{p \in D} w_p = 1. \quad (20)$$

2.7.3 Chromosome and Context Weights

Optionally, chromosome and context can be assigned aggregation weights:

$$\lambda_c \geq 0, \quad \sum_c \lambda_c = 1, \quad \mu_t \geq 0, \quad \sum_t \mu_t = 1. \quad (21)$$

Let $D_{c,t} \subseteq D$ be the subset of final DMPs on chromosome c and context t .

2.7.4 Global Group Log-Likelihood

The global weighted log-likelihood of sample i under group g is then defined as:

$$L_g(i) = \sum_c \lambda_c \sum_t \mu_t \left(\frac{1}{|D_{c,t}|} \sum_{p \in D_{c,t}} w_p \ell_{g,p}(y_{i,p}) \right), \quad (22)$$

where $|D_{c,t}|$ is the number of DMPs in chromosome c , context t . The factor $1/|D_{c,t}|$ can be used to prevent large blocks from dominating the score.

If chromosome/context weighting is not desired, a simplified variant is:

$$L_g(i) = \sum_{p \in D} w_p \ell_{g,p}(y_{i,p}). \quad (23)$$

2.7.5 Posterior Probabilities and Decision Rule

Assuming group priors $\pi_g = P(g)$, the posterior score for group g is:

$$S_g(i) = L_g(i) + \log \pi_g. \quad (24)$$

The posterior probability of group g for sample i is:

$$P(g | i) = \frac{\exp(S_g(i))}{\sum_{g'} \exp(S_{g'}(i))}. \quad (25)$$

The final classification is:

$$\hat{g}(i) = \arg \max_g S_g(i). \quad (26)$$

For two classes, {Healthy, Disease}, one can define the log-likelihood ratio:

$$\Delta(i) = S_{\text{Disease}}(i) - S_{\text{Healthy}}(i), \quad (27)$$

which can be used as a continuous risk score.

3 Results

This section is reserved for empirical evaluations of MethylPipeline, including:

- computational scalability (runtime, memory),
- accuracy of DMP detection,
- performance of the generative classifier (e.g., AUC, accuracy, balanced accuracy),
- comparison with baseline pipelines and classifiers.

4 Discussion

Here, one can discuss the implications of using Beta Mixture Models and generative classification, the advantages of working with centroids instead of full matrices, and the potential extensions of the framework (e.g., multi-class diseases, longitudinal data, integration with other omics layers).

5 Conclusion

MethylPipeline provides a scalable, statistically principled, and biologically interpretable framework for WGBS-based disease classification. By combining GPU-accelerated alignment, centroid-based aggregation, mixture modeling, and generative classification, the pipeline can handle large cohorts and exploit the full structure of methylation distributions at selected cytosine sites. The Beta Mixture-based classifier respects per-position uncertainty, multimodality, and biological context, offering a robust foundation for early detection and biomarker discovery.

Acknowledgments

The authors acknowledge the contributions of the broader Epimethyl Analytics team and collaborators involved in data generation, system administration, and scientific discussion.

References

A Additional Mathematical Details

This appendix can include:

- details on estimation of Beta parameters from centroid statistics,

- optimization details for fitting Beta Mixture Models,
- approximation methods for KL or JS divergences between mixtures,
- theoretical remarks on the properties of the weighted log-likelihood classifier.

B Notation Table

Table ?? summarizes the main notation used in the manuscript.

Symbol	Description
g	Group index (Healthy, Disease)
c	Chromosome index
t	Cytosine context (CG, CHG, CHH)
p	Genomic position (or position index)
i	Sample index
$mC_{i,p}$	Methylated read count for sample i at position p
$uC_{i,p}$	Unmethylated read count for sample i at position p
$y_{i,p}$	Methylation level of sample i at position p
$N_{g,p}$	Number of samples in group g with valid coverage at p
$\bar{y}_{g,p}$	Group-wise mean methylation at position p
$s_{g,p}^2$	Group-wise variance of methylation at position p
$D^{(0)}$	Preliminary set of differentially methylated positions (DMPs)
D	Final set of DMPs after mixture refinement
$f_{g,p}(y)$	Beta Mixture density for group g at position p
$\pi_{g,p,k}$	Mixture weight of component k for $f_{g,p}$
$\alpha_{g,p,k}, \beta_{g,p,k}$	Shape parameters of Beta component k
w_p	Total weight for position p
λ_c	Chromosome-level aggregation weight
μ_t	Context-level aggregation weight
$L_g(i)$	Global weighted log-likelihood for group g and sample i
$S_g(i)$	Posterior score for group g and sample i
$\Delta(i)$	Log-likelihood ratio (Disease vs Healthy) for sample i

Table 1: Summary of notation used in MethylPipeline.

C Multi-Class Extension of the Generative Classifier

This appendix describes the extension of the generative MethylClassifier to the multi-class setting in which several cancer types are modeled simultaneously. Let the set of biological groups be

$$\mathcal{G} = \{\text{Healthy}, \text{Cancer}_1, \dots, \text{Cancer}_K\},$$

where each Cancer_k corresponds to a specific tumor type or molecular subtype. All components of the model defined in the Methods section (centroids, mixture models, divergences, biological weights) extend naturally to this richer label space.

C.1 Per-Group Beta Mixture Models

For each group $g \in \mathcal{G}$ and each DMP position $p \in D$, we fit a Beta Mixture Model (BMM)

$$f_{g,p}(y) = \sum_{k=1}^{K_p} \pi_{g,p,k} \text{Beta}(y \mid \alpha_{g,p,k}, \beta_{g,p,k}), \quad (28)$$

exactly as in the binary case, with $K_p \in \{1, 2, 3\}$ typically determined by information criteria (AIC/BIC). Each cancer type obtains its own mixture parameters, allowing distinct spectrum shapes across groups.

Given an observed methylation level $y_{i,p}$ for sample i , the log-likelihood of group g at position p is defined as:

$$\ell_{g,p}(y_{i,p}) = \log \left(\sum_{k=1}^{K_p} \pi_{g,p,k} \text{Beta}(y_{i,p} \mid \alpha_{g,p,k}, \beta_{g,p,k}) + \varepsilon \right). \quad (29)$$

C.2 Global Weighted Log-Likelihood per Group

Let $w_p \geq 0$ denote the weight of position p , normalized to $\sum_{p \in D} w_p = 1$. The simplest (and most effective) multi-class global score is:

$$L_g(i) = \sum_{p \in D} w_p \ell_{g,p}(y_{i,p}), \quad (30)$$

although chromosome- and context-level weighting can also be included as in the main Methods section.

C.3 Posterior Distribution Over All Groups

Assign priors $\pi_g = P(g)$ (commonly uniform if no prevalence correction is desired). The posterior score is:

$$S_g(i) = L_g(i) + \log \pi_g. \quad (31)$$

A softmax over all groups gives the multi-class posterior:

$$P(g \mid i) = \frac{\exp(S_g(i))}{\sum_{h \in \mathcal{G}} \exp(S_h(i))}. \quad (32)$$

Thus the classifier returns a full probability vector:

$$(P(\text{Healthy} \mid i), P(\text{Cancer}_1 \mid i), \dots, P(\text{Cancer}_K \mid i)).$$

C.4 Global Cancer vs. Healthy Probability

Because all cancer subtypes are treated explicitly, the global cancer probability is obtained by collapsing the multi-class posterior:

$$P(\text{Cancer} \mid i) = \sum_{c=1}^K P(\text{Cancer}_c \mid i) = 1 - P(\text{Healthy} \mid i). \quad (33)$$

This definition requires no additional model. A binary decision may be made using a threshold τ :

$$\hat{g}_{\text{binary}}(i) = \begin{cases} \text{Cancer}, & P(\text{Cancer} \mid i) > \tau, \\ \text{Healthy}, & \text{otherwise.} \end{cases}$$

C.5 Interpretation

The model preserves the key advantages of the generative formulation:

- it uses the complete Beta Mixture structure per-position;
- it produces an interpretable probability per cancer type;
- “global cancer is obtained directly as $1 - P(\text{Healthy} \mid i)$;
- no separate binary model is needed;
- training is purely mixture-parameter estimation there is no iterative optimization across samples, making the method scalable.

C.6 Hyper-Parameters to Optimize

The full pipeline contains two natural phases—biomarker discovery and classification—each with its own hyper-parameters.

Phase 1: DMP Selection and Mixture Modeling

The following hyper-parameters influence which cytosine sites are selected:

- Divergence thresholds (Hellinger, JS, Jeffreys).
- Effect-size thresholds for mean or variance differences.
- q -value significance threshold.
- Maximum number of mixture components K_p .
- Model-selection criterion (AIC, BIC, ICL).
- Minimum coverage or robustness filters.
- Biological weighting scheme (promoter, exon, gene-priority weights).

These hyper-parameters determine the final marker panel D and the distribution models $f_{g,p}$.

Phase 2: Generative Classification

The classifier introduces additional hyper-parameters:

- Position weights w_p (statistical, biological, divergence-derived).
- Optional group-specific weights $w_{p,g}$.
- Chromosome weights λ_c and context weights μ_t .
- Priors π_g for the posterior computation.
- Threshold τ for global cancer detection.

Tuning these hyper-parameters governs:

- cancer vs. healthy sensitivity and specificity,
- subtype discrimination accuracy,
- stability of posterior probabilities across samples.

C.7 Summary

The multi-class generative classifier is a direct generalization of the binary model; no architectural changes are required beyond computing a likelihood per group and applying a multi-class softmax. The global cancer probability emerges naturally from the posterior, and the framework remains fully compatible with the per-chromosome, per-context centroid and mixture modeling workflow.

D Sample-Level Analysis Using Beta Mixture Models

This appendix describes how the fitted Beta Mixture Models (BMMs) for each differentially methylated position (DMP) can be used to analyze individual samples. The procedures described here support:

1. detection of substructure within a biological group (unsupervised clustering), and
2. statistical association of mixture components with external covariates such as age, BMI, race, and clinical variables.

All methods rely on the posterior membership probabilities of each mixture component for each sample and DMP.

D.1 Posterior Component Responsibilities

After the BMM has been fitted for group g and position $p \in D$, the mixture density takes the form

$$f_{g,p}(y) = \sum_{k=1}^{K_p} \pi_{g,p,k} \text{Beta}(y \mid \alpha_{g,p,k}, \beta_{g,p,k}), \quad (34)$$

with mixture weights $\pi_{g,p,k}$ and component parameters $(\alpha_{g,p,k}, \beta_{g,p,k})$.

For a given sample i with methylation value $y_{i,p}$ at position p , the posterior probability (“responsibility”) that the value arises from component k is:

$$\gamma_{i,p,k} = P(Z_{i,p} = k \mid y_{i,p}, g, p) = \frac{\pi_{g,p,k} \text{Beta}(y_{i,p} \mid \alpha_{g,p,k}, \beta_{g,p,k})}{\sum_{h=1}^{K_p} \pi_{g,p,h} \text{Beta}(y_{i,p} \mid \alpha_{g,p,h}, \beta_{g,p,h})}. \quad (35)$$

These satisfy:

$$\gamma_{i,p,k} \geq 0, \quad \sum_{k=1}^{K_p} \gamma_{i,p,k} = 1.$$

A hard assignment is defined by:

$$\hat{k}_{i,p} = \arg \max_k \gamma_{i,p,k}. \quad (36)$$

The collection $\{\gamma_{i,p,k}\}_{p \in D, k}$ represents the complete sample-level mixture fingerprint of sample i .

D.2 Aggregated Mixture Features for Each Sample

To compare samples, it is useful to aggregate responsibilities across DMPs. Define the global mixture usage score of sample i for component k as:

$$F_{i,k} = \frac{1}{|D|} \sum_{p \in D} w_p \gamma_{i,p,k}, \quad (37)$$

where w_p are the DMP weights introduced in the main Methods section. The vector

$$\mathbf{F}_i = (F_{i,1}, \dots, F_{i,K_{\max}})$$

acts as a low-dimensional “mixture signature” summarizing how sample i expresses the dominant methylation modes across all selected positions.

Region-specific signatures may also be constructed. For a genomic region $\mathcal{R} \subseteq D$:

$$F_{i,k}^{(\mathcal{R})} = \frac{1}{|\mathcal{R}|} \sum_{p \in \mathcal{R}} w_p \gamma_{i,p,k}. \quad (38)$$

These signatures enable fine-grained comparisons across promoter regions, pathways, chromosomes, or any biologically defined subset.

D.3 Clustering Samples Within a Group

Given mixture signatures $\mathbf{F}_i$, one may cluster samples within a single biological group g to identify substructure. Common choices include:

- k -means or Gaussian mixture clustering on $\mathbf{F}_i$,
- hierarchical clustering,
- density-based clustering (e.g., HDBSCAN),
- nonlinear embeddings (t-SNE, UMAP) applied to the matrix $\{\mathbf{F}_i\}$ prior to clustering.

Clustering based on mixture features often reveals:

1. subgroups of patients with distinct methylation modes,
2. potential molecular subtypes within a cancer group,
3. patient subgroups dominated by specific BMM components (e.g. hypermethylated vs. intermediate vs. hypomethylated modes).

D.4 Associating Mixture Components with Covariates

The responsibilities $\gamma_{i,p,k}$ and the aggregated features $F_{i,k}$ allow investigation of associations between methylation state patterns and external covariates such as age, BMI, tumor stage, race, or clinical outcomes.

Per-DMP Associations

At a fixed DMP p , associations may be tested using either:

- the hard labels $\hat{k}_{i,p}$ (categorical variable), or
- the soft assignments $\gamma_{i,p,k}$ (continuous variables).

Typical analyses include:

- regression of $\gamma_{i,p,k}$ on continuous covariates (age, BMI),
- ANOVA or KruskalWallis tests comparing $\gamma_{i,p,k}$ across categorical covariates (race, ethnicity),
- contingency tables for $\hat{k}_{i,p}$ vs. discrete covariates.

Global or Region-Specific Associations

Using the aggregated mixture signatures $F_{i,k}$ or $F_{i,k}^{(\mathcal{R})}$, one may fit models such as:

$$F_{i,k} = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{BMI}_i + \cdots + \varepsilon_i, \quad (39)$$

or compare distributions of $F_{i,k}$ across categories.

Such models reveal broad patterns such as:

- age-associated drift toward particular mixture components,
- demographic factors associated with methylation mode usage,
- clinical covariates affecting specific promoter- or pathway-level mixture structures.

D.5 Interpretation

The posterior component responsibilities $\gamma_{i,p,k}$ and their aggregated forms provide a structured bridge between

1. **distribution-level** methylation modeling (via BMMs),
2. **sample-level** phenotyping (clustering), and
3. **covariate-level** epidemiological or clinical analysis.

This representation retains the multimodal nature of methylation distributions, captures uncertainty in the latent component assignments, and provides a compact feature space for downstream statistical investigations.